

nProber III Platform

High productivity 7 nm capable nanoprobing system

nProber III is our most advanced SEM based nanoprobing platform. It provides electrical characterization and fault isolation capability for process development, device design debug and failure analysis down to the 7 nm node.

The platform combines high stability SteadFast CL™ nano-manipulators with the LEEN™ high-resolution scanning electron microscope column for precise, repeatable and stable probe placement on critical level features. The NM™ software control suite and eFast™ automation software provide integrated probe, SEM and parametric analyzer control for efficient operation.

The Thermo Scientific™ nProber III™ Platform includes eight (8) SteadFast CL low drift probe positioners and a high resolution sample stage. The sample stage can accommodate samples up to 50 mm or multiple smaller samples at once. The SteadFast CL nano-manipulators have been designed for low drift and repeatable motion to enable users to efficiently execute sensitive electrical tests on individual transistors. The nProber III platform has NM system control software allows users to setup key SEM parameters and measurement test plans in a single unified environment that improves operational efficiency and reduces system complexity. eFast automation provides a step by step workflow that guides the user through common nanoprobing tasks. The closed loop (CL) feedback design of the SteadFast CL nano-manipulators enables the automation of probe alignment and cross cleaning thereby improving ease of use.

nProber III platform is the first nanoprobing system to use the LEEN SEM column specifically designed to meet key nanoprobing requirements. The LEEN column operates at high resolution down to 200 eV with a $\pm 100 \mu\text{m}$ field of view (FOV), reducing ebeam impact on sensitive materials.

nProber III platform includes a small volume load lock for fast sample exchange. An advanced anti-contamination system, the Optimizer® (US Patent 5,510,624) is integrated into the load lock to remove contaminants on the sample surface. The nProber III platform can be configured with an optional *in-situ* sample touch up tool for the final step in sample preparation thereby reducing the need for iterative sample prep steps.

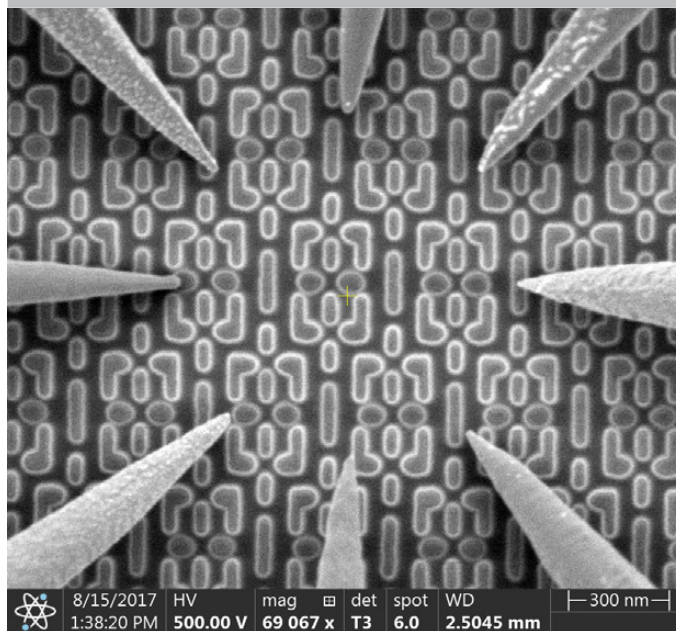
Key benefits

Platform stability, fast, repeatable probe motion control and low drift rates for stable electrical measurements on advanced design node devices

Integrated system operation, guided work flows and software control of measurements, probes and SEM setup for efficient operation

SEM column designed for nanoprobing

High resolution at low landing energies with large FOV



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HV
500.00 V

mag
69 067 x

det
T3

spot
6.0

WD
2.5045 mm

300 nm

Nanoprobing applications

DC Measurement:

DC Measurement Software (DCM) allows users to easily create, save and run electrical tests through the main nProber III user interface. The DC characterization system includes an industry standard nine (9) channel parametric analyzer and is designed for low-noise measurements. Test results (tables and curves), automatically calculated device parameters, test settings and images automatically captured during the test sequence are saved together and accessed using DataViewer software. The DataViewer software can also be installed on multiple customer computers for offline analysis, manipulation and extraction for report generation.

Pulsed I-V Measurements

Pulsed I-V measures transistor I-V curves in response to high speed signals sent to the transistor gate. This is used to localize failures associated with resistive gates that can be missed by DC I-V measurements. The Test and Measurement (TAM) software integrates the parametric analyzer together with a high speed pulser and oscilloscope allowing the user to easily create, save test set ups and run pulsed I-V tests through the nProber III NM control suite.

C-V Characterization

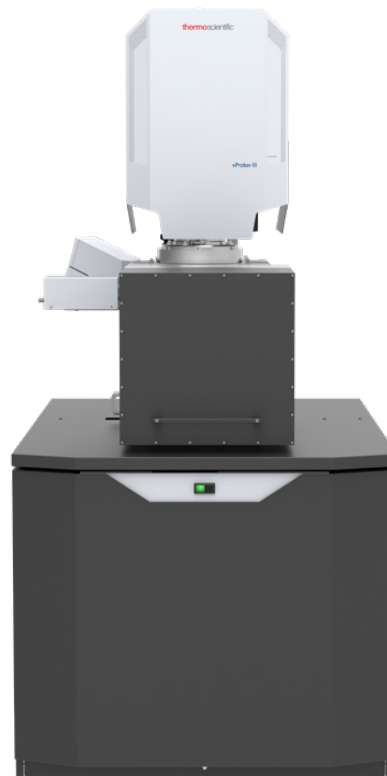
The C-V application in the Test and Measurement (TAM) software integrates and controls an LCR Meter allowing the user to easily create, save test set ups and measure device gate capacitances through the nProber III platform user interface.

E-Beam Characterization (EBC):

EBC expands the nProber III platform capability to localize high resistance faults in device metallization through Electron Beam Absorbed Current Imaging (EBAC) and Electron Beam Induced Current (EBIC). EBAC and EBIC use the scanning electron beam at increased current and landing energy to inject current into buried metal structures multiple levels deep to localize the high resistance faults. The EBC package includes software to create and manipulate the EBC signal that can then be overlaid on the SEM image of the sample to identify the location of the electrical fault.

E-Beam Induced Resistive Change (EBIRCH)

Patented EBIRCH™ capability localizes low resistance faults in device metallization and transistor structures by detecting changes in the electrical properties of a fault when a high energy electron beam is scanned across the fault. EBIRCH operation is programmed with the EBIRCH Application in the Test and Measurement (TAM) software. The EBIRCH signal is digitally mapped to the simultaneously collected SEM image to provide visual confirmation of the fault location.



Summary

nProber III is our most advanced SEM based nanoprobing platform, provides electrical characterization and fault isolation capability to support process development, device design debug and failure analysis down to the 7 nm node.

The nProber III platform has SteadFast CL nano-manipulators along with the LEEN high-resolution scanning electron microscope column and advanced control software allow for precise, repeatable and stable probe placement on critical level features.

Find out more at thermofisher.com/EM-Sales

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